

Design and Characterization of Imbalanced Mach-Zehnder Interferometers in Silicon Photonics.

Jirayu Mongkolkiattichai

I. INTRODUCTION

Photonic Technology relies on the manipulation of light in waveguide structures, which consist of a high-refractive-index (core) surrounded by a lower-refractive-index (cladding) medium. The properties of a waveguide can be engineered in proper ways, for example, with respect to dispersion, mode, and polarization. Many applications are realized using photonics, not only in telecommunication but also in sensing and quantum optics. A crucial tool in these applications is interferometry. Interferometers come in several forms, including the Michelson, Mach-Zehnder, and Fabry-Perot designs. In this work, I will focus on the Mach-Zehnder interferometer (MZI).

II. THEORY

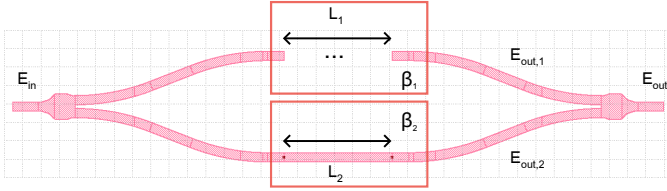


Fig. 1. **Schematic of a MZI.** Light is split by a Y-branch and propagates along the upper and lower arms with different lengths. The light is then recombined by a Y-branch, resulting in a transmission profile.

At the right end of the waveguides shown in Fig. 1, the electric field propagating along the upper and lower arms can be written as

$$E_{out,i} = \frac{E_{in}}{\sqrt{2}} e^{-i\beta_i L_i - \frac{\alpha_i}{2} L_i}, \quad (1)$$

where E_{in} denotes the input field to the device, β_i represents the propagation constant and α_i is the propagation loss [1].

From the lecture, I learn that the output field from the last Y-brach is a combination of the fields from upper and lower arms, divided by the square root of two, i.e.,

$$E_{out} = \frac{1}{\sqrt{2}} (E_{out,1} + E_{out,2}). \quad (2)$$

The output field intensity is proportional to $|E|^2$, and I can express it as follows

$$I_{out} = \frac{I_{in}}{2} [1 + \cos(\beta_1 L_1 - \beta_2 L_2)], \quad (3)$$

where I treat $\alpha = 0$ as a special case in which propagation loss is absent. For an imbalanced interferometer with identical waveguides ($\beta_1 = \beta_2 = \beta$), I can simplify Eq. 3 to

$$I_{out} = \frac{I_{in}}{2} [1 + \cos(\beta \Delta L)]. \quad (4)$$

As can be seen, Eq. 4 is a sinusoidal function, and it is interesting to find the spacing between adjacent peaks, known as the free spectral range (FSR), which can be obtained by expressing

$$\text{FSR} = \Delta\lambda = \lambda_{m+1} - \lambda_m \quad (5)$$

where m is the spectral order. From the mode calculation, the effective index n is wavelength-dependent. As a result, the propagation constant is no longer constant across all wavelengths [2]. In fact, I can apply the first-order Taylor expansion of β such that

$$\beta_{m+1} \approx \beta_m + \frac{d\beta}{d\lambda} \Delta\lambda. \quad (6)$$

I rearrange this expression to obtain

$$\Delta\beta = \beta_m - \beta_{m+1} \approx -\frac{d\beta}{d\lambda} \Delta\lambda. \quad (7)$$

Let's consider $d\beta/d\lambda$ where $\beta = 2\pi n/\lambda$, the first derivative results in

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda} \frac{dn}{d\lambda} + 2\pi n (-\lambda^{-2}) = -\frac{2\pi}{\lambda^2} \left(n - \frac{dn}{d\lambda} \lambda \right). \quad (8)$$

Finally, the FSR depends on three parameters: wavelength, path length difference, and the group index.

$$\text{FSR} = \Delta\lambda = \frac{\lambda^2}{\Delta L (n - \lambda \frac{dn}{d\lambda})} = \frac{\lambda^2}{n_g \Delta L}. \quad (9)$$

III. MODELING AND SIMULATION

Here, I consider a waveguide geometry with a height of 220 nm and a width of 500 nm, and TE polarization is assumed. The fundamental TE mode in the waveguide is calculated using Lumerical MODE as shown in Fig. 2. The software

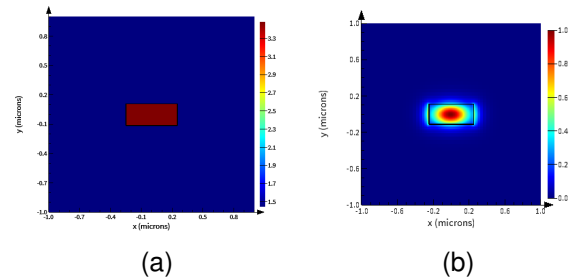


Fig. 2. **Fundamental Mode Simulation.** (a) Waveguide schematic, (b) 2d intensity profile of the TE₀₀ mode.

allows us to obtain several physical parameters including effective index (n_{eff}), group index (n_g), loss, and effective area. It is interesting to study the variation of the effective index

and the group index as a function of wavelength (Fig. 3). The variation of n_{eff} is approximately 0.1, whereas n_g varies by about 0.01 for a wavelength range spanned by 100 nm centered at 1550 nm. The wavelength-dependent effective index can be

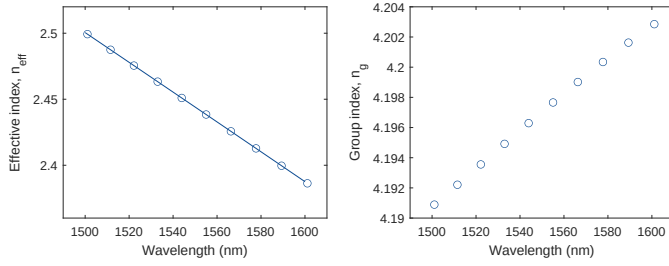


Fig. 3. **Effective and group indices.** (left) Effective index n_{eff} as a function of wavelength λ . The solid line indicates the compact model. (right) Group index n_g as a function of wavelength λ .

understood using a compact model for the waveguide, which I express as a polynomial function. Here, I use a quadratic polynomial. By fitting the simulated data points in Fig. 3(a), I obtain $n_{\text{eff}}(\lambda)$ as follows:

$$n_{\text{eff}} = 2.4441 - 1.1290(\lambda - 1.55) - 0.0391(\lambda - 1.55)^2. \quad (10)$$

This dispersion of the indices governs the wavelength-dependent phase accumulation in the MZI which contains several components. To understand its behaviors, I analyze each component by calculating the transfer function. In practice, I input a constant optical power, sweep the wavelength, and measure the gain. This can be performed straightforwardly using Lumerical INTERCONNECT. As shown in Fig. 4, the Y-branch exhibits an approximately constant gain of about -3 dB, whereas the grating-grating coupler has a full width at half maximum (FWHM) of approximately 50 nm. The purpose of having two grating-grating couplers with different waveguide lengths is to measure a propagation loss (Fig. 4(d)).

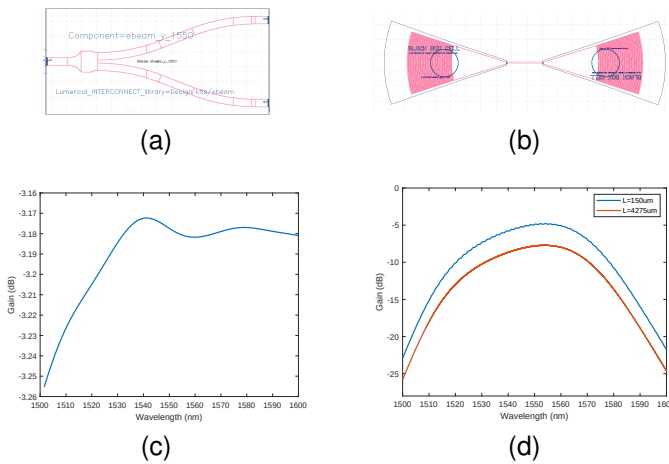


Fig. 4. **Transfer functions.** (a,c) Y-branch and its transfer function. (b,d) Grating-grating coupler and its transfer function for a short waveguide (150 μm) and a long waveguide (4275 μm).

By constructing the MZI from available components in the ebeam PDK, I prepared a system to run a full simulation (Fig. 5).

dL (μm)	calculated FSR (nm)	simulated FSR (nm)
50	11.4	11.2
100	5.7	5.6

TABLE I
MZI PARAMETERS.

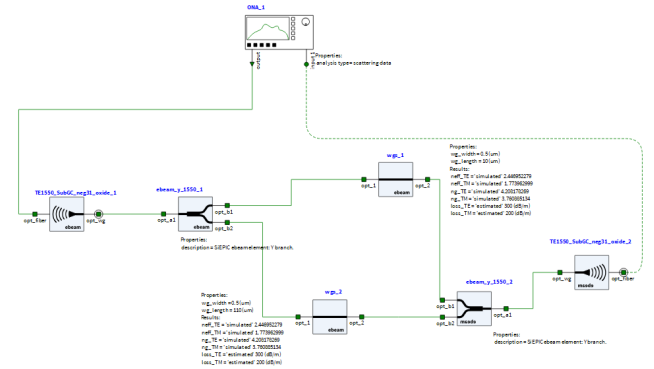


Fig. 5. **Schematic of MZI in Lumerical INTERCONNECT.**

I verify Eq. 9 by constructing Table I for the designed MZIs. Both calculated FSR and simulated FSR are nearly identical, with only a slight difference in the first decimal place. The simulation allows us to obtain the transfer function as demonstrated in Fig. 6. In the experiment, I will measure the transmission as a function of wavelength. I can extract the peak-to-peak spacing and determine the corresponding wavelength range which yields the FSR. The measured FSR is the substituted into to Eq. 9, given the known path length difference between the MZI arms. The group index can then be directly obtained. The designed layout is illustrated in Fig. 7.

IV. FABRICATION

My designed layout was merged into the main layout and it was submitted for fabrication at the Washington Nanofabrication Facility at the University of Washington. This facility provides rapid prototyping for passive silicon photonic components utilizing high-resolution electron beam lithography (EBL) and plasma etching. The fabrication sequence is summarized in the following steps (Fig. 8) [3]:

- 1) **Substrate Selection:** The process starts with a piece of a silicon-on-insulator (SOI) wafer, typically featuring a 220 nm silicon device layer and a 3000 nm buried oxide layer.
- 2) **Resist Application:** A negative hydrogen silsesquioxane (HSQ) resist is spin-coated onto the chip and then baked on a hotplate for a few minutes.
- 3) **EBL Exposure:** The pattern is written directly into the resist using a JEOL JBX-6300FS EBL system.
- 4) **Development:** After exposure, the resist is developed by immersing the sample in a tetramethylammonium hydroxide (TMAH) solution for a few minutes.
- 5) **Rinse and Dry:** The chip is cleaned with a 60-second deionized water rinse and a 10-second isopropanol rinse before being blown dry with nitrogen.
- 6) **Plasma Etching:** Pattern transfer is achieved by removing silicon from unexposed areas using chlorine-based

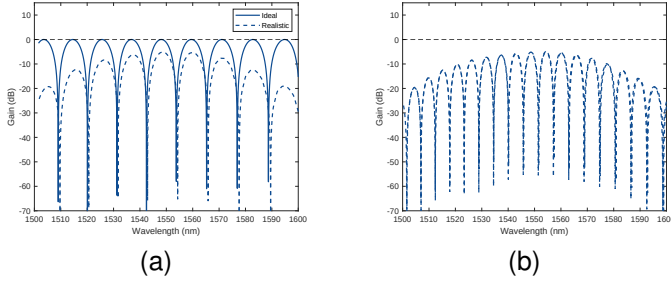


Fig. 6. **Transfer function of MZI.** (a) $dL = 50\mu\text{m}$ and (b) $dL = 100\mu\text{m}$. The solid line indicates an ideal case where no loss occurs and bandwidth is independent. The dashed line represents a realistic case that includes loss and wavelength dependence.

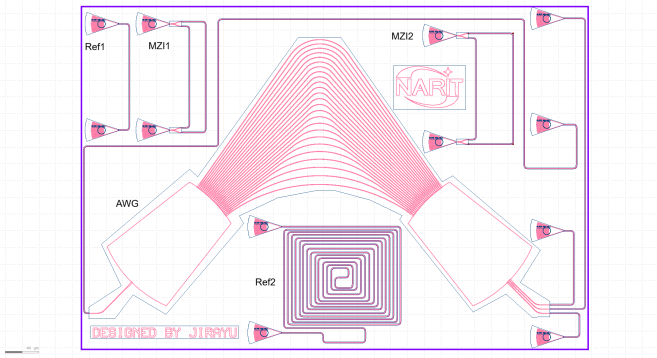


Fig. 7. **Layout of the designed devices.** Ref1 and Ref2 are reference waveguides; MZI1 and MZI2 are Mach-Zehnder interferometers; and AWG is an arrayed waveguide grating.

inductively coupled plasma (ICP) etching in an Oxford Plasmalab system.

- 7) Oxide Deposition: A protective oxide cladding layer, typically 2-3 μm thick, is deposited over the waveguides using PECVD to prevent mechanical damage and environmental contamination.
- 8) Chip Dicing: Finally, the original substrate is diced into individual chips.

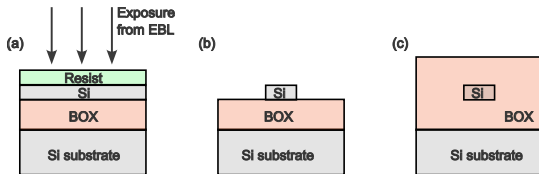


Fig. 8. **Fabrication process flow of the SOI structure.** (a) Electron-beam lithography exposure defines the pattern in the resist layer on top of the Si device layer. (b) Pattern transfer into the Si layer through etching, forming the waveguide structure on the buried oxide (BOX). (c) Final structure after resist removal, and oxide cladding is deposited, therefore showing the patterned Si device layer on the BOX and Si substrate.

V. EXPERIMENTAL DATA AND ANALYSIS

The measurement was performed by Iris stage and Scylla stage. For a completeness of dataset, I chose Iris stage dataset. It is worth noting that, according to Fig. 7, only the reference waveguides and Mach-Zehnder interferometers are analyzed in this work, while the AWG is left for future study.

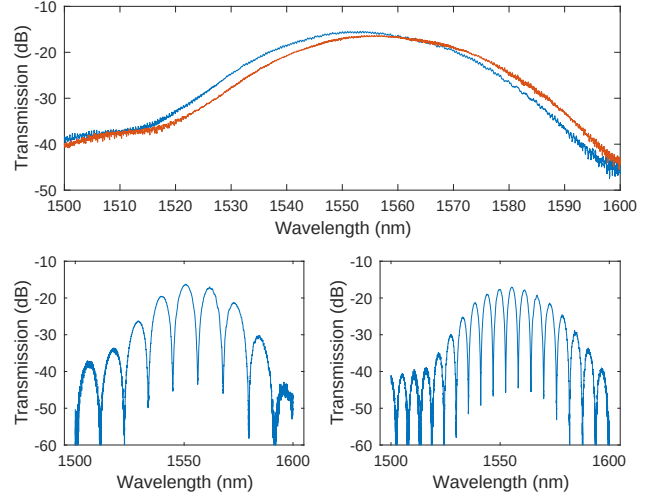


Fig. 9. **Measurement of reference waveguides and MZIs.** (Top) Reference waveguides with lengths of 150 μm (blue) and 4275 μm (red). (Bottom) Mach-Zehnder interferometers with $dL = 50\mu\text{m}$ (left) and $dL = 100\mu\text{m}$ (right).

To begin data analysis, I firstly load measured spectral data and converts the wavelength into SI units using MATLAB. It then performs baseline correction by fitting the raw transmission data [Fig. 9 (bottom)] to a 4th-order polynomial and subtracting this fitted background, effectively flattening the spectrum and isolating the interference fringes. The data is later restricted to a specific wavelength range (1.52–1.58 μm) to remove noisy edge regions.

Using a theoretical MZI transmission model, I performed a nonlinear least-squares fit to match the experimental data by setting effective refractive index up to second orders, propagation loss, offset as free parameters. The quality of the fit was evaluated using the coefficient of determination (R^2). The best fit is shown in Fig. 10 (top).

Here, a corner analysis was performed to account for deviations in the waveguide geometry, which leads to variations in the effective refractive index and group index. The analysis of the MZIs defined in Table I is shown in Fig. 10 (bottom). The extracted group index agrees within the bounds predicted by the corner analysis, implying that the observed variations originate from dimensional nonuniformity of the waveguide across the Si substrate.

In Fig. 9 (top), I planned to analyze the propagation loss constant and insertion loss using a simple model, $\text{Loss} = A \times L + B$, where A is the propagation loss per unit length, L is the length of a reference waveguide, and B is the insertion loss due to grating coupling on both sides. Unfortunately, the designed racetrack waveguide is not long enough and it also intersects at 1560 nm, so no conclusion can be drawn from this measurement.

VI. CONCLUSION

In this study, imbalanced Mach-Zehnder interferometers were successfully designed and fabricated on a silicon-on-insulator platform to extract physical parameters. Experimental results confirmed that the extracted group index (n_g) aligns

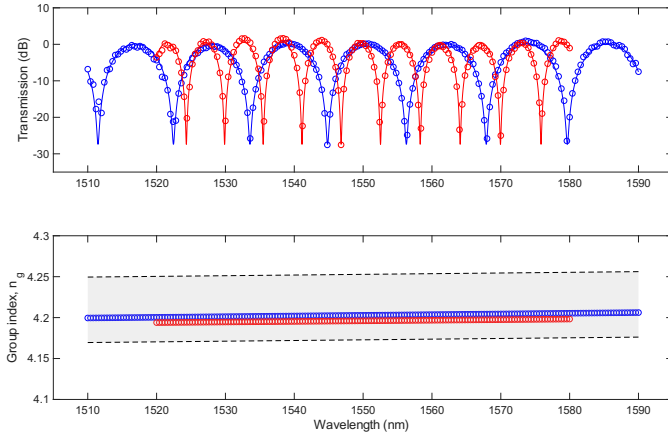


Fig. 10. **Measurements of the MZIs.** (Top) Transfer functions. Blue and red represent MZIs with ΔL equal to 50 and 100 μm , respectively. Dots indicate measured data with downsampling applied for clarity, while solid lines show fits to the model. (Bottom) Group index as a function of wavelength. The gray region represents tolerances calculated from a corner analysis, with heights varied from 215.3 nm to 223.1 nm and widths varied from 470 nm to 510 nm.

closely with theoretical models and fits within the bounds predicted by corner analysis. This consistency suggests that observed variations in the group index are primarily attributable to dimensional non-uniformities introduced during the electron-beam lithography and etching process. These results validate the MZIs as a robust diagnostic tool for monitoring fabrication consistency, with future work focusing on the characterization of the integrated arrayed waveguide grating.

VII. ACKNOWLEDGMENTS

I acknowledge the Center for Optics and Photonics, National Astronomical Research Institute of Thailand (Public Organization) for supporting my participation in the Silicon Photonics course. I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design*. Cambridge, U.K.: Cambridge University Press, 2015.
- [2] L. Chrostowski, "The free spectral range (FSR) of the imbalanced interferometer," presented at UBCx Phot1x course, EdX, Jan 2026.
- [3] R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," *J. Vacuum Sci. Technol. B* 29, 06F309, 2011.